No Hydrogen at the Wow! Signal's Velocity in Its Two Candidate Fields

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Abstract

A reanalysis of the Ohio State data of 1977 places the Wow! signal at 1420.726 MHz, a velocity of $-74\,\mathrm{km\,s^{-1}}$ in the local standard of rest, in one of two fields at declination $-26^\circ57' \pm 20'$, and reads it as a flare of stimulated 21 cm emission from a cold hydrogen cloud, which must then lie on one of those lines of sight at that velocity. The HI4PI survey shows none. Averaged over $10'$ at each field the brightness at $-74\,\mathrm{km\,s^{-1}}$ is 0.010 and 0.003 K against a noise of 0.019 K, and a line one channel ($1.29\,\mathrm{km\,s^{-1}}$) wide is limited to a column of $1.4 \times 10^{17}\,\mathrm{cm^{-2}}$ at 3σ, while each field's own emission peaks at $+3$ to $+4\,\mathrm{km\,s^{-1}}$ at 26 K. Within each field's $\pm20'$ the brightest pixel between -79 and $-69\,\mathrm{km\,s^{-1}}$ reaches 2.8 times its noise of 0.040 K, none of the 8800 pixels within 3° of either field reaches five times its own noise there, and the mean of those pixels shows only the smooth wing of the local emission, 0.014 K, with no feature at the velocity. A cloud filling HI4PI's $16.2'$ beam at the signal's velocity lies in neither field; one of angular size θ smaller than the beam is limited to $1.4 \times 10^{17}(16.2'/\theta)^2\,\mathrm{cm^{-2}}$ in a single channel, a limit that only interferometric 21 cm data at arcminute resolution can lower.

1 Introduction

On 1977 August 15 the Big Ear radio telescope of Ohio State University recorded a narrowband source near the 21 cm line that rose and fell as the sky drifted through its beam and was never seen again, the Wow! signal. Searches of the same sky since, among them with the Very Large Array [1] and at the Australian radio telescopes [2], found nothing. Arecibo Wow! I proposed an astrophysical origin: a transient brightening of a cold hydrogen cloud by stimulated emission, triggered by a passing burst of radiation [3]. Arecibo Wow! II reanalysed the archival Ohio data and revised the signal's properties: a frequency of 1420.726 MHz, a velocity of $-74\,\mathrm{km\,s^{-1}}$ in the local standard of rest, and one of two fields, at right ascensions $19^\mathrm{h}25^\mathrm{m}02^\mathrm{s}$ and $19^\mathrm{h}27^\mathrm{m}55^\mathrm{s}$ and declination $-26^\circ57'$ within $\pm20'$ [4].

A stimulated-emission flare needs hydrogen at the signal's velocity on the line of sight, and the HI4PI survey [5] maps the 21 cm line over the whole sky at $16.2'$ resolution. This note reads HI4PI at the two fields.

2 Data and method

HI4PI combines the Effelsberg-Bonn H I Survey and the Galactic All-Sky Survey [6], which covers these fields, into spectra on a HEALPix grid of $3.4'$ pixels with channels of $1.29\,\mathrm{km\,s^{-1}}$ in the local standard of rest. For each field we average the spectra of the pixels within $10'$ of its centre, 38 and 40 pixels, and read the brightness at $-74\,\mathrm{km\,s^{-1}}$, the brightest channel within $\pm5\,\mathrm{km\,s^{-1}}$ of it, and the noise from the emission-free channels beyond $\pm200\,\mathrm{km\,s^{-1}}$. A line one channel wide at 3σ has, if optically thin, a column $N = 1.823 \times 10^{18} \int T_B\,dv\,\mathrm{cm^{-2}}$ [7]. Each pixel within the field's

Table 1: HI4PI at the two fields of Arecibo Wow! II. T_{-74} is the brightness of the 10′ average at $-74\,\mathrm{km\,s^{-1}}$, σ its noise, $T_{\max}$ the brightest channel within $\pm5\,\mathrm{km\,s^{-1}}$, $N_{3\sigma}$ the column of a one-channel line at 3σ, and v_{peak} and T_{peak} the field's own emission.

Field	l	b	T_{-74} (K)	σ (K)	$T_{\max}$ (K)	$N_{3\sigma}$ ($\mathrm{cm^{-2}}$)	v_{peak}, T_{peak}
$19^{\mathrm{h}}25^{\mathrm{m}}02^{\mathrm{s}}$	11.62°	−18.82°	0.010	0.019	0.032	1.4×10^{17}	$+4.0\,\mathrm{km\,s^{-1}}$, 25.8 K
$19^{\mathrm{h}}27^{\mathrm{m}}55^{\mathrm{s}}$	11.87°	−19.41°	0.003	0.019	0.047	1.3×10^{17}	$+2.7\,\mathrm{km\,s^{-1}}$, 26.4 K

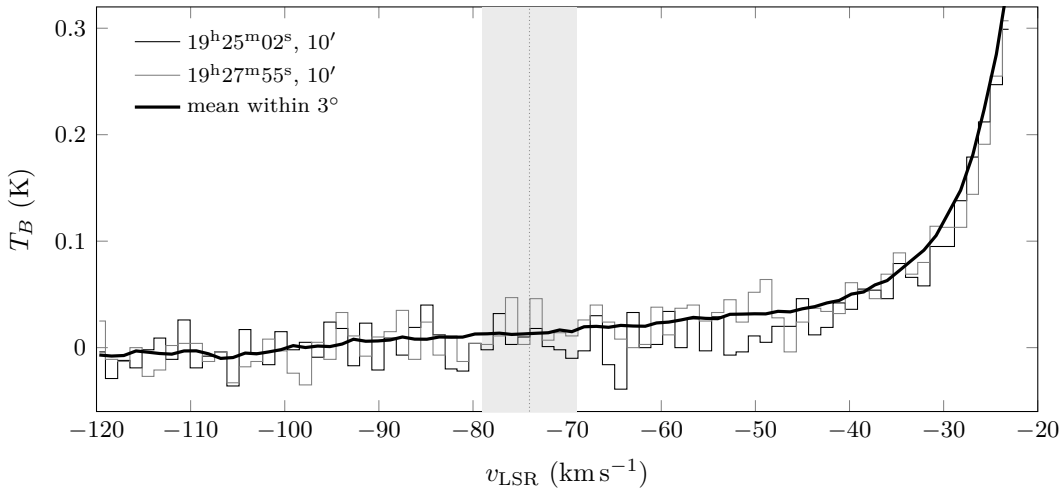


Figure 1: HI4PI spectra of the two fields averaged within 10′, and the mean of the 8826 pixels within 3° of the first, from −120 to $-20\,\mathrm{km\,s^{-1}}$; the grey band is −79 to $-69\,\mathrm{km\,s^{-1}}$ and the dotted line the signal's $-74\,\mathrm{km\,s^{-1}}$. The fields' own emission, off the plot, peaks at +3 to $+4\,\mathrm{km\,s^{-1}}$ with 26 K.

uncertainty, ±3 s of right ascension and ±20′ of declination, is read on its own between −79 and $-69\,\mathrm{km\,s^{-1}}$, and every pixel within 3° is tested there against five times its own noise. The mean spectrum of those pixels shows any diffuse gas at the velocity.

3 Results

Table 1 gives the two fields. Neither holds emission at the signal's velocity: the 10′ averages are 0.010 and 0.003 K at $-74\,\mathrm{km\,s^{-1}}$ against a noise of 0.019 K, the brightest channels within $\pm5\,\mathrm{km\,s^{-1}}$ reach 1.6 and 2.5 times the noise, and a one-channel line is limited to 1.4×10^{17} and $1.3\times10^{17}\,\mathrm{cm^{-2}}$. Each field's hydrogen lies near zero velocity, peaking at +4.0 and $+2.7\,\mathrm{km\,s^{-1}}$ with 26 K (Figure 1). Inside each field's uncertainty, 21 and 20 pixels, the brightest at the velocity holds 0.106 and 0.113 K, 2.7 and 2.8 times the single-pixel noise of 0.040 K, as the largest of some 160 independent channels does by chance. None of the 8826 and 8836 pixels within 3° reaches five times its own noise between −79 and $-69\,\mathrm{km\,s^{-1}}$. Their mean spectrum, with a noise of 0.0026 K a channel, carries the wing of the local emission falling smoothly from 0.03 K at $-50\,\mathrm{km\,s^{-1}}$ through 0.014 K at the signal's velocity to the baseline near $-100\,\mathrm{km\,s^{-1}}$, with no feature at $-74\,\mathrm{km\,s^{-1}}$.

4 Related work

The Wow! signal's sky has been revisited with the Very Large Array [1] and at the Australian radio telescopes [2] in search of a recurrence, and Arecibo Wow! I and II [3, 4] proposed and refined an astrophysical origin in cold hydrogen. We find no earlier reading of 21 cm surveys at the revised fields and velocity.

5 Conclusion

HI4PI shows no hydrogen at the Wow! signal's revised velocity in either of its candidate fields: no emission above its noise in the $10'$ averages, no pixel within 3° at five times its own noise, and only the smooth wing of the local emission in the region's mean. A cloud that fills the $16.2'$ beam at $-74\,\mathrm{km\,s^{-1}}$ is excluded to a column of $1.4 \times 10^{17}\,\mathrm{cm^{-2}}$ in a single channel. A stimulated-emission source that the Big Ear's beam could catch need not fill HI4PI's, and a compact cloud of angular size θ escapes these data below $1.4 \times 10^{17}(16.2'/\theta)^2\,\mathrm{cm^{-2}}$, about $4 \times 10^{19}\,\mathrm{cm^{-2}}$ for $\theta = 1'$. Interferometric 21 cm maps of the two fields at arcminute resolution, which the declination of -27° puts within reach of the southern arrays, would test the cloud the hypothesis needs.

Acknowledgements

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